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Generative ecodesign for mechanical products: A design workflow

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ABSTRACT

Harnessing advancements in artificial intelligence, generative design holds great potential to support designers in their ecodesign efforts by enabling them to explore design solutions beyond the limits of their imagination and expertise. However, a systematic literature review on the application of generative design in ecodesign reveals a clear underrepresentation, highlighting a missed opportunity in the field. To bridge this gap, a seven-component generative ecodesign workflow for mechanical products was developed. This workflow combines generative design algorithms, typically used for geometry lightweighting, with life cycle thinking. It facilitates the generation, evaluation, and identification of design solutions by considering the design tri-factor: material choice, manufacturing process, and geometry. This represents the first reported product ecodesign tool to integrate generative design with ecodesign principles while simultaneously addressing all three elements of the design tri-factor. To showcase its utility, environmentally optimal design alternatives were created for a mountain bicycle's handlebar stem.

1. Introduction

Design decisions made during the product design stage can significantly affect up to 80% of its environmental impact (Komisjon et al., 2012; Ellen MacArthur Foundation). Given this influence, it is important to incorporate environmental consideration in design decisions at this early stage to achieve the greatest environmentally sustainable improvements. To do so product designers turn to ecodesign.

Developed in the 1990s by the design community, ecodesign was conceived to optimise a product's environmental performance (Dusch and Moultrie). Ecodesign broadens the traditional paradigm of product design by integrating environmental considerations alongside aesthetic and functional performance concerns (Karlsson and Luttropp, 2006/01). With an intense focus on reducing global greenhouse gas emissions from industrial activities to curb climate change, ecodesign is invaluable as the successful application of ecodesign can lower a product's production carbon footprint and serve as a pre-emptive measure to reduce the carbon emission of new designs.

Designers play an essential role in implementing ecodesign during the design stage (Schäfer and Löwer, 2021). To support designers in their ecodesign endeavours, various ecodesign tools have been developed (Schäfer and Löwer, 2021; Lee et al., 2023/10). Ecodesign tools, particularly those used to generate design solutions, are predominantly prescriptive, featuring checklists, guidelines, diagrams, and methodologies (Willocx and Duflou, 2021/01). Prescriptive ecodesign tools offer designers generic guidelines on approaches to reduce the environmental impact of their designs. These ecodesign tools do not present specific design solutions but rely on the designer's interpretation of the guidelines to generate solutions, leading to varied results. Consequently, a common critique of these ecodesign tools centres on the lack of direct support provided to designers and their dependence on the designer's expertise, experience, and creativity to formulate design solutions (Lee et al., 2023/10; Willocx and Duflou, 2021/01; Lofthouse, 2006/01).

With recent developments in artificial intelligence, generative design has emerged as an invaluable tool for designers. Particularly in the area of lightweighting, generative design generates innovative geometries

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that reduce the material used to create the design while considering constraints such as loading conditions, design boundaries, and input materials. The prospect of using generative design for ecodesign is appealing, as it could provide direct design solutions without solely relying on the designer's expertise.

However, current generative design tools are predominantly designed to focus on performance or aesthetics rather than environmental sustainability. A review of generative design tools, elaborated upon in the literature review, reveals that despite the proliferation of generative design tools, existing tools are not specifically set up for product ecodesign. For example, in the application of generative design for lightweighting, current tools focus primarily on minimising design weight to meet design requirements without considering the environmental sustainability of the design. Although lightweighting is an ecodesign strategy, its implementation does not always improve the environmental sustainability of the base design when viewed from a life cycle perspective (Lee et al., 2023/01). While using less material can reduce the emission associated with raw material extraction, the lightweight structure could be more complex to fabricate, resulting in higher emissions in other life cycle stages.

To bridge the gap identified, a generative ecodesign workflow for mechanical products was developed in this work. The seven-component workflow integrates the concept of ecodesign with geometry generative design. The objective of the workflow is to generate an environmentally sustainable optimal design solution that fulfils the design objective from the starting design. In the context of this work, the design solution will comprise the design tri-factor: the choice of material, the choice of manufacturing process, and design geometry. In this work, the environmental sustainability of the designs will be quantified by the greenhouse gas potential, measured by the equivalent carbon dioxide (kg CO_2).

The paper proceeds as follows: Section 2 presents a literature survey, Section 3 outlines the generative design workflow, and Section 4 provides a case study demonstration, illustrating the implementation of the workflow using the handlebar stem of a mountain bicycle. Section 5 offers a discussion, and Section 6 concludes the paper.

2. Literature review

2.1. Generative design

Generative design can be used by designers for design explorations or design optimization (Ballegaard et al., 2021). Upon reviewing various literature sources, it is notable that subtle nuances exist in the definitions of generative design, depending on the context and application (Süße et al., 2022/01). Despite these differences, certain common themes emerge. Fundamentally, generative design is described as an iterative design process that generates outputs that fulfil predefined criteria or constraints through computational methods (Gradišar et al., 2022; Krish, 2011/01). These methods may include the use of software (Bhilare et al., 2023), computational models (Gradišar et al., 2022) artificial intelligence methods, and algorithms (Buonamici et al., 2020), such as evolutionary algorithms (Zhang et al., 2020/01), to solve design problems.

2.1.1. Application of generative design in product design

In product design, generative design is commonly used for geometry exploration or optimization, particularly in structural applications. In these contexts, generative design generates design solutions through an iterative optimization process that minimises the objective function until an optimal solution is generated (Buonamici et al., 2020; Pilagatti et al., 2023/06; Yi et al., 2021/01). The objective function can include factors such as the design's loading condition, material properties, and/or process constraints (Pilagatti et al., 2023/06).

Although generative design was not originally developed for sustainability improvement, its application in geometry exploration or optimization for lightweighting has emerged as a viable ecodesign strategy. In lightweighting, the objective function typically aims to minimise a design's mass, thereby potentially reducing material usage and, consequently, lowering material emissions. However, this technique primarily focuses on mass minimisation without accounting for the life cycle emissions of the design. For instance, one lightweighting approach involves replacing solid structures with lattice structures to achieve equivalent mechanical performance using less material. While this can reduce material emissions due to decreased material usage, the increased fabrication complexity of such designs can lead to higher production emissions due to factors such as longer production times and the use of more energy-intensive manufacturing methods. As a result, the cumulative emission of the lightweight design could, in some cases, exceed those of the initial design. Therefore, while the application of generative design holds promises for ecodesign improvement, it should be approached with caution.

2.2. Applications of generative design in ecodesign

2.2.1. Systematic literature review methodology

To acquire a deeper insight into the existing literature that integrates generative design in ecodesign, a systematic literature review, as outlined by Denyer and Tranfield (2009), was undertaken. The search and filter process is summarised in Fig. 1. To initiate the review process, a search string was entered into the Scopus database. At the time of the review, 320 search results were obtained, covering the publication period from 1982 to 2023. On cursory screening where duplicate results and titles of conferences and workshops are removed, 274 articles remain. Next, the title and the abstract are screened followed by the assessment of the contents of the article. In this series of filtering, articles that are not written in English, do not relate design to environmental sustainability, describe design science, describe the role of social/user behaviour in ecodesign, and only reports on case study implementation will be removed. At the end of the filtering process, 17 article remain.

2.2.2. Review synthesis and observations

Based on the review of the shortlisted articles (Table 1), it is evident that literature discussing the application of generative design for ecodesign predominantly focuses on architecture, with approximately 82% (14 articles) in this category, and the remaining 18% relating to product design (3 articles). A common sustainability strategy observed in architecture ecodesign using generative design involves optimising building designs based on climatic conditions to reduce energy consumption for lighting, heating, and cooling applications. In some cases,

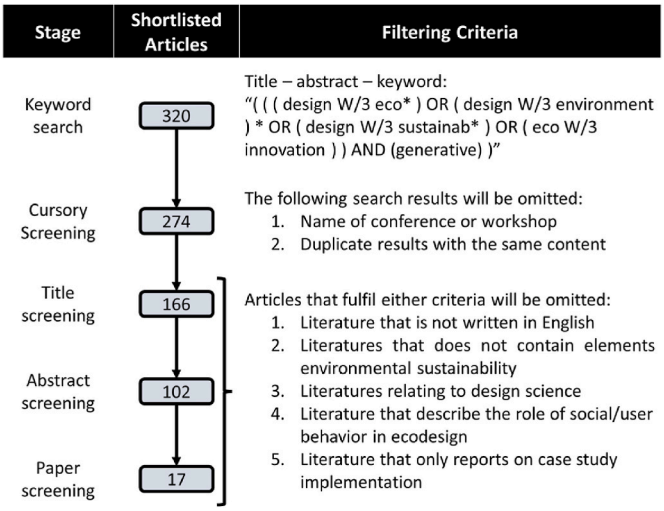


Fig. 1. Schematic overview of the literature selection process.

Table 1
Description of shortlisted article from the systematic literature review search.

Idx	Title	Application	Description	Sustainability Angle	Ref
1	Using BIM to Facilitate Generative Target Value Design for Energy Efficient Buildings	Architecture (Building Design)	Proposed generative target value design framework that combines generative design analysis and target value design to generate energy-efficient building designs. Primarily used for solar and energy analysis, the framework optimises the building design through minimisation of the annual energy consumption and solar incident on the building façade while maximising the surface area for rooftop solar panels.	Reduce the environmental footprint of the building by optimising the building design to minimise energy consumption while maximising the area of renewable energy usage.	Bhilare et al. (2023)
2	Generative design for more economical and environmentally sustainable reinforced concrete structures	Architecture (Building Design)	Presents a simplified generative design approach for concrete beam design that reduces material usage and carbon dioxide emission.	Reduce the environmental footprint of the construction of reinforced concrete beams by reducing unnecessary usage of material.	(Alsakka et al., 2023/02)
3	GENE_ARCH: An evolution-based generative design system for sustainable architecture	Architecture (Building Design)	Discuss GENE_ARCH, an evolution-based generative design system developed to help architects create energy-efficient sustainable architectural design solutions.	Generate energy-efficient building shapes through the consideration of climatic data to minimise the building heating load while maximising daylighting.	Caldas (2006)
4	Towards Integration and Hybridization in Urban Generation: An extendable urban generative system for better natural ventilation	Architecture (Building Design)	Present an urban generative design framework to generate urban massing with optimised outdoor natural ventilation.	Reduce the environmental footprint of the building by minimising the energy consumption required for cooling by maximising natural ventilation.	Zhang et al. (2023)
5	Generative Sustainable Performance Design Exploration Applied to Greek Urban Typologies	Architecture (Building Design)	Developed a parametric model for the Greek Urban Typology, the Polykatoikia, to identify sustainable design using generative design.	Reduce the environmental footprint of buildings by minimising the energy consumption of buildings by minimising the building's summer solar load while maximising its winter solar load.	(Christodoulou and Symeonidou, 2023/06)
6	BIM-Based Building Geometric Modeling and Automatic Generative Design for Sustainable Offsite Construction	Architecture (Building Design)	Proposed a 3D building geometric modelling method that uses generative design to generate possible geometric variation for precast to optimise carbon footprint and construction cost to support sustainable and efficient offsite construction.	Reduce the environmental footprint of offsite construction for precast buildings by reducing the quantity of material used.	Gan (2022)
7	Toward a national life cycle assessment tool: Generative design for early decision support	Architecture (Building Design)	Developed a framework for a national life cycle assessment prototype tool to generate an optimal design for the exterior walls design of residential buildings considering Egyptian constraints.	Evaluate and select a generated design that has the lowest embodied material carbon.	(Hassan et al., 2022/07)
8	GoDesign: A modular generative design framework for mass-customization and optimization in architectural design	Architecture (Building Design)	Present a modular generative design framework for the built environment. Environmental sustainability objectives can be raised and incorporated in the planning stage of the framework.	Optimise the sustainability of building design by optimising the building design to reduce the required energy load of the building.	Azadi and Nourian (2021)
9	Shaping of curvilinear steel bar structures for variable environmental conditions using genetic algorithms—moving towards sustainability	Architecture (Building Design)	Presents an optimization method that applies generative shaping tools to obtain environment-adaptive and structurally effective curvilinear steel bar structures to support sustainable designing.	Optimise the sustainability of curvilinear steel bar structures by reducing the steel required.	Dzwierzynska (2021)
10	Performance in generation: An automatic generalizable generative-design-based performance optimization framework for sustainable building design	Architecture (Building Design)	Proposed a generalizable generative-design-based performance optimization framework that combines generative design with performance-based design techniques to facilitate green building design.	Reduce the building's environmental footprint by reducing the energy required for lighting, heating, and cooling applications.	(Ji et al., 2023/11)
11	Buildable performance envelopes	Architecture (Building Design)	Describe a computational tool that generates an optimal buildable performance envelope considering the climate and usage context of the site. The tool uses a voxel visualization approach to analyse climatic variables which will inform the generation of design that aims to maximise energy efficiency of the building.	Generate energy-efficient design through the consideration of climatic data to minimise the building heating load while maximising daylighting.	Thariyan et al. (2017)
12	An interactive and generative eco-design tool for architects in the sketch phase	Architecture (Building Design)	Present a generative ecodesign software that can be used for the early stages of an architectural project. Allow designers to generate eco-efficient building designs for a built environment.	Generate building design considering energy or environmental performance.	(Marsault and Torres, 2019/11)

(*continued on next page*)

Table 1 (*continued*)

Idx	Title	Application	Description	Sustainability Angle	Ref
13	Framework for increasing sustainability of factory systems by generative layout design	Architecture (Interior Design)	Proposed an approach for sustainability-based factory layout planning using generative design by extending the optimization criteria and restrictions considered in the existing generative design framework developed for layout planning.	Reduce the environmental footprint by generating a factory layout that reduces energy usage.	(Süße et al., 2022/01)
14	Self-organising floor plans in care homes	Architecture (Interior Design)	Presents an optimising approach for spatial layouts that increase floor plan efficiency based on a genetic algorithm.	Optimise the layout to reduce the energy consumption of the space.	Carta et al. (2020)
15	Sustainable component design using generative tools for additive manufacturing	Product design	Show that generative design can increase a designer's productivity in a sustainable manner. As it allows them to explore a large number of design options for a single part in a short period of time without having to fabricate the prototypes for testing.	Achieve sustainability by reducing the energy and material waste from testing prototypes in different usage scenarios.	Cucu et al. (2021)
16	Data-Driven Generative Design Integrated with Hybrid Additive Subtractive Manufacturing (HASM) for Smart Cities	Product Design	Discuss the integration of generative design with a hybrid additive subtractive manufacturing approach.	Advocate the use of hybrid additive subtractive manufacturing instead of conventional manufacturing to reduce energy and material usage. The integration of generative design offers weight optimization which could reduce the material required and potentially energy required for certain applications (E.g., component on a drone).	Dilibal et al. (2021)
17	A case study on the part optimization using eco-design for additive manufacturing based on energy performance assessment	Product design	Use the energy performance assessment based ecodesign for additive manufacturing method to predict the energy performance of additive manufacturing to fabricated industrial parts generated by generative design.	Evaluate and select the generated design considering the energy required to produce the designs.	(Yi et al., 2021/01)

it is also used to enhance renewable energy generation by maximising the solar irradiance on the building rooftop for solar panel deployment. This focus on energy consumption primarily addresses the usage phase of an architecture project's life cycle emissions. Although less common, literature on sustainability strategies that impact the raw material extraction stage has also been identified. In these works, generative design is used to generate designs that reduce the embodied carbon associated with the components/material used in construction (Alsakka et al., 2023/02; Gan, 2022; Hassan et al., 2022/07; Dzwierzynska, 2021). The emphasis on the usage phase in architecture ecodesign can be seen as a strategic focus on the life cycle stage with the most significant environmental impact. Given the long lifespan of architectural projects, energy usage during a building's usage phase typically dominates the emission profile, outweighing emissions from other life cycle phases.

In product design, due to the dearth of literature on the subject, no clear trends were identified in the reviewed works. Among the studies examined, generative design was employed to reduce resource use in design exploration (Cucu et al., 2021), minimise material usage in products (Dilibal et al., 2021), and lower energy consumption in production processes (Yi et al., 2021/01). While generative design has been applied for lightweighting, an ecodesign strategy, it is often not explicitly classified as an ecodesign tool, as it can, in some cases, leads to higher cumulative emissions, as discussed in the previous section. Consequently, it is frequently categorized as a functional tool for weight reduction rather than an ecodesign tool. However, there remains an opportunity for generative design use for lightweighting to evolve into a genuine ecodesign tool. The underrepresentation of generative design tools for product ecodesign, coupled with its latent potential for environmental sustainability improvement through lightweighting, presents a significant opportunity for further development. This work aims to seize that opportunity by developing a generative ecodesign workflow for mechanical products, building on lightweighting principles to improve environmental sustainability.

It is important to note that the systematic literature review conducted in this work offers a preliminary exploration of generative design in ecodesign applications. It is not intended to be an indictment of the potential for generative design to support ecodesign efforts. Instead, it highlights the need for deeper investigation into various generative design methodologies and algorithms to uncover their latent potential for ecodesign applications.

3. Generative design workflow

To adapt generative design for use in the ecodesign of mechanical products, a generative ecodesign workflow consisting of seven components was developed, as illustrated in Fig. 2. In this approach, the design tri-factor is considered concurrently due to the interdependencies between the design variables and their influence on the product life cycle (Appendix A). This workflow was created with careful attention to the intervariable relationships within the design tri-factor, enabling

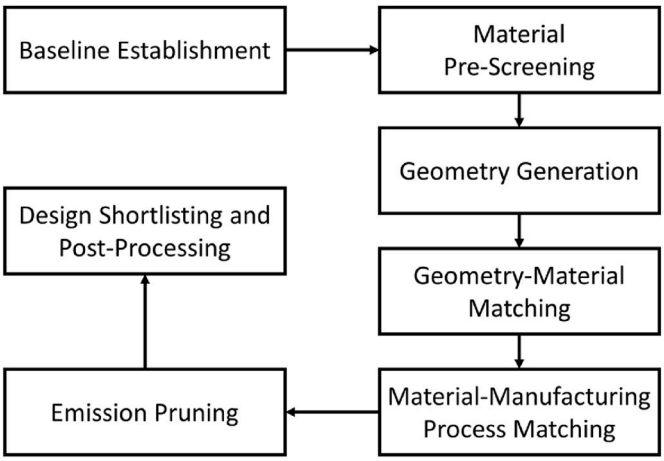


Fig. 2. Generative ecodesign workflow.

designers to generate alternative design solutions that concurrently address these factors.

Developed to enhance the environmental sustainability of designs, the workflow is intended for use after the design conception stage, once a baseline design has been established. In this work, the environmental sustainability of the designs will be quantified by the greenhouse gas potential, measured in terms of equivalent carbon dioxide (kg CO_2). The remainder of this sub-chapter will explore the functionalities of each component in the generative ecodesign workflow.

3.1. Step 1: baseline establishment

The baseline establishment component aims to define baseline requirements that guide the generation and selection of environmentally sustainable design solutions. This involves considering the design's functional requirements, design conditions, and the baseline design. In this work, baseline requirements can be broadly categorized into three categories: fixed baselines, dynamic baselines, and emission baselines.

The fixed baseline delineates design requirements whose fulfilment is contingent upon a predetermined baseline value or condition that remains unaffected by changes in the design solutions. This means that the baseline value or condition remains consistent and unaltered regardless of modifications to the design solutions. Consequently, design variables that are immutable, where the baseline choice is predetermined, will also be listed as fixed baseline requirements (e.g., manufacturability of material for a predetermined manufacturing process).

The dynamic baseline defines requirements whose fulfilment depends on predetermined baseline values or conditions that are influenced by changes in the design solutions. For instance, when considering the factor of safety in a design with design geometry as a design variable, the baseline value required for the tensile strength to meet the factor of safety changes with the choice of geometry, even though the factor of safety value itself remains constant. It is important to note that the classification of a requirement as a fixed or dynamic baseline is not rigid. The categorization of a requirement as fixed or dynamic depends on the context of the functional requirement and the design variables being considered.

Finally, the emission baseline defines the maximum allowable emission level for a design solution to qualify as a sustainable alternative to the baseline design. The emissions of a design can be described by the amalgamation of emission categories, which reflect the emissions of each life cycle stage. Each emission category is composed of multiple emission constituents, which reflect the emissions from the activities within each life cycle phase. The workflow draws inspiration from the comparative life cycle assessment (LCA) approach to evaluate the emissions of the generated design against the baseline design. This approach creates a permission structure to omit emission constituents and categories whose contributions to the product's emissions remain unaffected by modifications to any of the design tri-factors.

To achieve this, designers must consider the emission ontology for the product they are designing before computing the emission baseline. Fig. 3 provides an example of a generic emission ontology for a mechanical product without omitting any emission constituents and categories. After the designers have decided on the emission ontology the baseline emission can be computed using equation (1). In the equation, n represents the total number of emission constituents, EC, in the emission ontology.

$$\begin{matrix}\text{Emission}\\ \text{Baseline}\end{matrix} = \sum_{X=1}^{n} \begin{matrix}\text{Emission}\\ \text{Constituent}\\ X_{\text{Baseline}}\end{matrix} \tag{1}$$

For a more comprehensive description of the emission constituents, readers are encouraged to refer to Appendix B, which features descriptions of the emission constituents and high-level equations that describe how they can be computed.

3.2. Step 2: material pre-screening

The material pre-screening component screens and shortlists materials that have the potential to meet the functional requirements of the design. This component gathers inputs from pertinent fixed baseline requirements to guide the screening process. It is noteworthy that not all fixed baseline requirements are directly related to material selection, necessitating the consideration of only relevant requirements within this component.

3.3. Step 3: geometry generation

The geometry generation component is responsible for generating alternative design geometry options. Fig. 4 illustrates the inputs, outputs, and the sequential process of the component. In the redesign of mechanical products, typical requirements guiding the generation of alternative design geometries include the baseline design, its loading condition, design boundaries, and input materials. To ensure the viability of the alternative geometries generated and to streamline subsequent analysis in the workflow, the component employs a three-step process: alternative geometry generation, post-processing and analysis, and manufacturing process assignment.

In the initial step of the three-step process, a variety of alternative design geometries is generated using a generative design algorithm. Given that the goal of the workflow is on sustainability improvement,

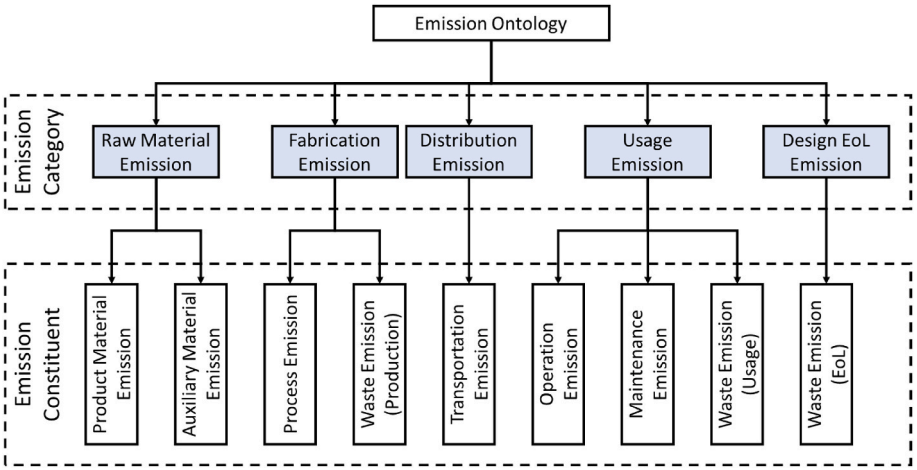


Fig. 3. Structure of a generic emission ontology.

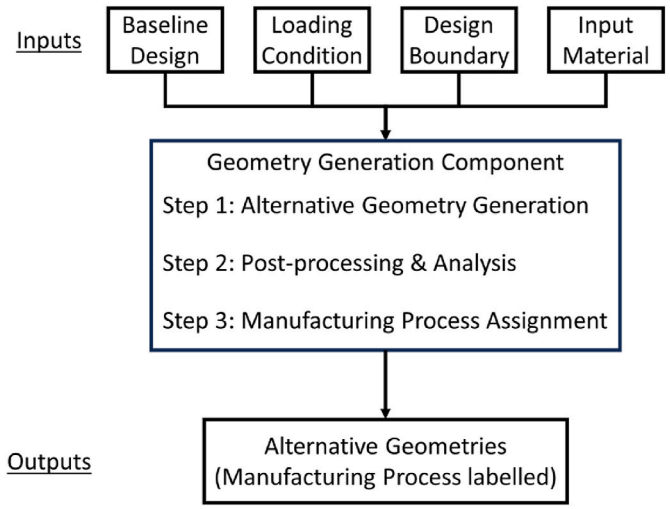


Fig. 4. Overview of the geometry generation component for generating alternative geometries for mechanical products.

minimising the design mass is recommended as an objective function for the generative design operations.

To facilitate the generation of diverse design geometries using generative design, designers can conduct parallel runs of the algorithm with different starting inputs derived from the modifications of the baseline inputs associated with any of the design tri-factors. For the design geometry, modifications to the baseline geometry can create different starting geometries, triggering the generation of diverse design solutions. At this stage, it is unnecessary to ensure that the starting geometry fulfils the loading conditions. Examples of modifications to the baseline geometry include adding perforations, varying dimensions, or removing volume based on finite element analysis or topology optimization results (nTopology). When considering the manufacturing process, designers can expand the design solutions by generating geometries that account for different objective functions based on process constraints. For the choice of material, designers can employ sampling methods to extract materials with varying properties from the list of pre-screened materials. Inputting every single material from the pre-screen materials is not advisable, as this would exponentially increase the number of simulations required and drastically drive up the lead time and computational cost of the generative design process.

Following the generation of alternative geometries, the second step in the three-step process involves the post-processing and analysis of these generated geometries. In this step, each generated geometry undergoes a visual inspection and any necessary repairs to ensure geometry continuity (e.g., no hanging volumes) while preserving key features. Subsequently, the geometries undergo mechanical analysis to ensure compliance with fixed and dynamic baseline requirements. The analysis results will be used to update the dynamic baseline unique to each generated geometry. These geometry-specific dynamic baselines are then utilized in the geometry-material matching component.

In the last step of the three-step process, the generated geometries will be inspected and assigned manufacturing processes that are suitable for their fabrication. This step may be omitted if the alternative geometry is generated with adherence to a defined manufacturing process as its objective function and when the designer has no desire to explore the possibility of the geometry's manufacturability under another manufacturing process.

3.4. Step 4: geometry-material matching

Following the generation of alternative design geometry, the geometry-material matching component becomes active in the workflow to curate different geometry-material pairings. This component utilizes the geometry-specific dynamic baselines established in the geometry generation component, to identify materials that enable the geometry-material pairing to meet the functional requirements. Conceptually, this component can be seen as a continuation of the material pre-screening component, as it exclusively considers materials shortlisted from that component to curate each geometry-material pairing.

3.5. Step 5: material-manufacturing process matching

The next component in the workflow is the material-manufacturing process component. In this component, the manufacturing process labelled for each generated geometry, from the geometry generation component, is used to filter out materials that are not suitable for the labelled manufacturing process. For geometries labelled to be fabricable by multiple manufacturing processes, each geometry-material pairing will be evaluated for each manufacturing process listed. Should the material be suitable for the labelled manufacturing process that is being investigated a geometry-material-manufacturing process design solution will be generated.

It should be noted that for geometries labelled to be fabricable by multiple manufacturing processes, the geometry-material pairing is not required to be suitable for all the labelled manufacturing processes. The geometry-material pairing will only be removed for the manufacturing processes that the material is not suitable to be fabricated with.

3.6. Step 6: emission pruning

The emission pruning component evaluates and benchmarks the environmental sustainability of the generated alternative design solutions against the emission baseline established in the baseline establishment component. Symbolically, this component marks the conclusion of the design exploration phase within the generative ecodesign workflow, signalling a transition towards convergence and the identification of a final solution. The component utilizes the concept of relative sustainability to facilitate the converging process by pruning design solutions that have a comparatively higher emission compared to the baseline emissions (Hrivnak, 2007). Therefore, the emission constituents considered for the alternative design solution should be the same as those used in the computation of the baseline emission. By similarity, the emission category and emission constituent considered for the generated design would be the same as those considered in the computation of the emission baseline.

3.7. Step 7: Design shortlisting post-processing

The final step in the generative ecodesign workflow is the design shortlisting and post-processing component. This component comprises two parts: the final shortlisting of design solutions and the post-processing of design solutions. There is flexibility in the sequence of operations for this component, prioritisation or exclusion of either the shortlisting or post-processing part of the component is entirely up to the designers.

3.7.1. Final shortlisting of design solutions

To prioritize environmental sustainability, design solutions should be selected based on their respective emission reductions compared to the baseline design. The reduction factor, RF, of each design solution can be computed using Equation (2). This reduction factor allows for the ranking of the design solutions, enabling designers to establish a cutoff for shortlisting.

$$\mathrm{RF} = \frac{\mathrm{Emission}_{\mathrm{Baseline}} - \mathrm{Emission}_{\mathrm{Design\ Solution}}}{\mathrm{Emission}_{\mathrm{Baseline}}} \times 100\% \quad (2)$$

3.7.2. Post-processing of design solutions

Post-processing involves reviewing and potentially modifying the shortlisted design solutions to ensure manufacturability. While the geometry generation component already includes a "post-processing & analysis" step aimed at ensuring geometry continuity and the retention of important features, the focus in this component is on ensuring that the dimensions and features are ready for production. Designers may choose to post-process every shortlisted design solution from the previous component or focus on the top-ranked design solutions shortlisted using the reduction factor.

Besides ensuring manufacturability, designers can choose to optimise the design solutions by regenerating the design geometry guided by the parameters of the selected design solution. For the chosen design solution to be optimised, its design geometry, material, and manufacturing process will be used as input to guide the generation of the new design geometry. Subsequently, the newly optimised design solution will undergo post-processing and analysis to ensure that it complies with the design requirements and is manufacturable. Due to the need for additional resources, it is advisable to reserve this process for a limited number of designs that exhibit promising potential, as determined by the designers. It should be noted that post-processing of the design solutions is not required for all design scenarios.

4. Case study implementation: handlebar stem

To demonstrate the implementation of the generative ecodesign workflow, the handlebar stem of a mountain bicycle will serve as a case study. The baseline geometry of the handlebar stem used in the case study is modified from the one sourced from GrabCAD (Prentice), as depicted in Fig. 5. For the baseline design, the handlebar stem is assumed to be fabricated from milling (Our) of a rectangular Aluminium 6061 T6 blank (Jiang et al., 2020). In this case study's design scenario, only the handlebar stem of the mountain bike is being considered for modification. Consequently, connector features of the geometry must be conserved to ensure compatibility with other components of the mountain bike.

In the case study, the workflow will be supported by a curated material database comprising 50 materials, excluding the baseline material. The material list can be found in Appendix C, where each material was assigned a material index. This index will be used as an identifier for their respective material in place of the material name for the remainder of the section; the baseline material will be identified by the identifier MBL. Pertinent information regarding the material's physical properties, mechanical properties, processing properties, durability, and emissions (Embodied, processing, and EoL) is supplemented using data from Ansys Granta Selector (I. ANSYS) unless stated otherwise. Regarding the manufacturing process, the workflow will consider additive, subtractive, and formative manufacturing processes.

4.1. Baseline Establishment

This sub-section will commence by outlining the functional requirements of the handlebar stem. Subsequently, the baseline requirements of the case study will be established. The baseline established in this component will serve as foundational criteria that will be used in subsequent components in the workflow.

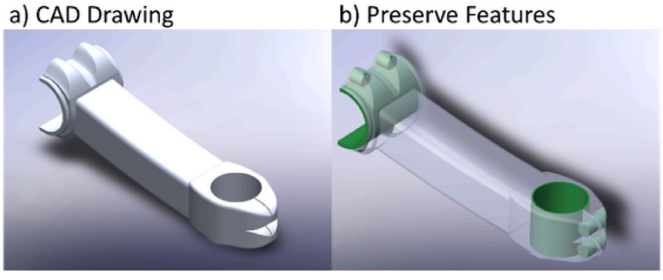


Fig. 5. CAD drawing of a mountain bike handlebar stem. Original CAD drawing sourced from GrabCAD (Prentice).

4.1.1. Functional requirements

Designed for installation on a mountain bike, the handlebar stem must meet functional requirements relating to its usage environment and safety. Firstly, as it is intended for outdoor use, the handlebar stem must be weatherproof to withstand exposure to the elements. Additionally, adherence to safety standards is essential to ensure user safety (Jiang et al., 2020). This study will utilize the tests outlined in ISO 4210–5:2023 to guide the component's safety standards, mandating a factor of safety of at least 1 for the component. ISO 4210–5:2023 provides comprehensive guidance on tests designed to ensure the durability and strength of each part of the bicycle, thereby ensuring the highest level of safety in bicycle operation (International Organization for Standardization, 2023a; Véle et al., 2023). For this case study, particular attention will be given to the static test, with a specific focus on the forward bending test concerning the design of the handlebar stem as illustrated in Fig. 6 outlined in ISO 4210–5:2023. The standard consists of two test methods, the description of the test method as well as the requirements for test fulfilment to the test is compiled in Table 2.

4.1.2. Fixed baseline

For the case study, fixed baselines are derived to address the functional requirements of the usage environment and the geometric constraints. A total of five fixed baseline requirements are identified for this case study. To ensure that the design solution is weatherproof, the material of the design solution must be durable against water and ultraviolet radiation, and it must retain its form (not melt) at an ambient temperature of 41 °C (Brocherie et al., 2020/04). The next fixed baseline requirement pertains to the preservation of specific geometry features based on the design scenario. To ensure compatibility with other components of the mountain bike, different regions of the baseline geometry are labelled (Moreno Nieto and Moreno Sánchez, 2021). Regions of the baseline geometry that come in contact with other components are designated as "preserve geometry" and will remain unmodified. These preserved geometry regions of the baseline design are highlighted in green as illustrated in Fig. 5b. The fifth baseline requirements stem from the requirements of the stage 1 forward bending test, which mandates

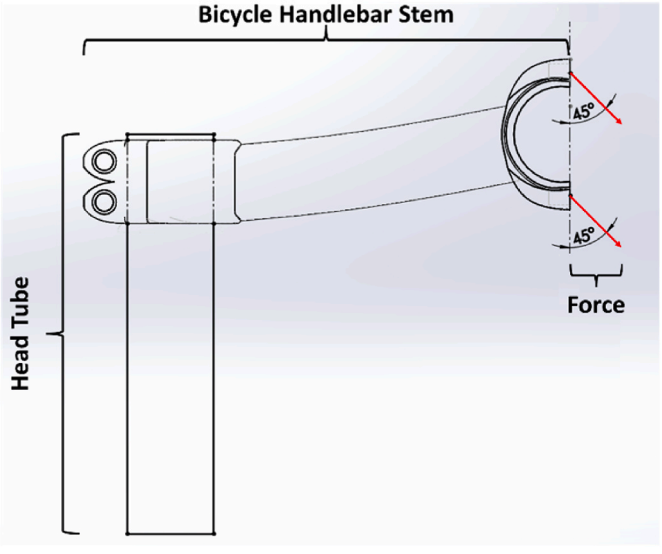


Fig. 6. Loading conditions of forward bending test for bicycle handlebar stem defined by ISO 4210-5-2023 (International Organization for Standardization, 2023a).

Table 2
Test method for stage 1 and 2 of the forward bending tests for the handlebar stem of a mountain bicycle. *Test method adapted from ISO 4210–5:2023 (International Organization for Standardization, 2023a) and ISO 4210–2:2023 (International Organization for Standardization, 2023b) respectively.

	Test Conditions	Requirements for test fulfilment
Stage 1	Apply a force of 1600N in the direction as shown in Fig. 6 and maintain the force for 1 min.	No visible cracks or fractures with permanent deformation measure <10 mm in the direction of the test force.
Stage 2	Apply a progressively increasing force in the direction as shown in Fig. 6 until the force reaches 2600 N or until the stem deflects ≥50 mm. Maintain the force for 1 min if the stem does not or continues to yield.	No visible cracks or fractures.

that the handlebar stem experiences deformation of less than 10 mm in the direction of the test force (1600N).

4.1.3. Dynamic baseline

The case study features a dynamic baseline specifically addressing the safety-related functional requirements of the handlebar stem. This involves ensuring that the handlebar stem has a factor of safety of at least 1.5. The evaluation of this factor of safety will be based on the loading conditions described by the stage 2 test method of the forward bending stress.

To meet the test requirements (Table 2), the handlebar stem must not exhibit visible cracks or fractures upon completion of the test. Therefore, the material's ultimate strength will be employed to compute the factor of safety (Autodesk Inca). Equation (3) provides the computation method for the factor of safety, which will be applied in this case study.

$$\text{Factor of Safety} = \frac{\text{Material Ultimate Strength}}{\text{Maximum Von Misses Stress}} \quad (3)$$

4.2. Emission baseline

In the emission baseline, the emission ontology of the case study must first be developed. Table 3 presents the emission constituents influenced by the design tri-factor for this case study. Among the emission categories, distribution emission and usage emission were found to be minimally affected by modifications to any of the design tri-factors.

In this case study, the alternative design is required to perform the same function as the baseline design. Given the similarity in functional requirements and the minimal resource input associated with the usage phase of a bicycle handlebar stem, the usage emissions for the baseline and alternative designs are expected to remain identical and unaffected by modifications in design variables. As for distribution emissions, they are more closely related to the distribution network than to the product design itself. Since the design tri-factor does not influence the distribution network, these emissions were excluded from the analysis. Based on these considerations, it was concluded that the design tri-factor would not significantly impact the usage and distribution phases of the product life cycle in this case study, justifying their exclusion from the emission baseline. However, it is important to acknowledge that in other case studies, the design tri-factor might have a more pronounced effect on these phases. Consequently, the exclusion of emission constituents and categories should be evaluated case-by-case.

From the emission constituent identified relevant equations are retrieved (Appendix A) to compute the emission baseline. Table 4 compiles the value of the variables collected from simulations and databases used to compute the emission baseline. The detailed computation of the emission baseline can be found in Table 5. Through this computation, the baseline emission for the case study is determined to be 15.32 kg CO_2.

Table 3
Influence of design variables on the emission constituents for a mechanical product.

Emission Category	Emission Constituent	Design Variable		
		Material	Manufacturing Process	Design Geometry
Raw Material Emission	Product Material Emission	✓		✓
	Auxiliary Material Emission	✓	✓	
Fabrication Emission	Process Emission	✓	✓	✓
	Waste Emission (Production)	✓	✓	
Distribution Emission	Transportation emission			
Usage Emission	Operation emission			
	Maintenance emission			
	Waste emission (Usage)			
Design EoL Emission	Waste Emission (EoL)	✓		✓

Table 4
Emission variable of baseline design solution.

Variable	Quantity	Unit	Remarks	Ref
Design Volume	107021.56	mm^3	Volume of baseline design	–
Blank Volume	415678.70	mm^3	Starting blank use to create the design	–
Density	2710	kg/m^3	Density of Aluminium 6061 T6	(I. ANSYS)
Embodied Carbon ofMaterial	8.85	kg CO_2/kg	Typical grade Aluminium 6061 T6	
Manufacturing Emission Factor	2.67	kWh/kg	Fine machining of Aluminium 6061 T6 in the unit of per weight of material removed.	
EoL Emission factor of Material	2.63	kg CO_2/kg	Carbon footprint from the recycling of Aluminium 6061 T6.	
Embodied Carbon of Energy	0.465	kg CO_2/kWh	Embodied carbon of electricity production in Singapore (Based on 2022 values)	(Energy Market Authority of Singapore)

4.3. Material pre-screening

In this case study, the curated material database will undergo pre-screening based on the fixed baseline requirement related to the functional need for the design solution to be weatherproof. This necessitates

Table 5
Computation of emission constituent for the baseline emission.

Emission Category	:	Raw Material Emission

The baseline design's raw material emissions consist of the product material emissions and the auxiliary material emissions. To compute either emission constituent, the mass of the material used in the baseline design must first be determined. Given that the baseline design is fabricated through milling, the mass of the auxiliary material is calculated by considering the volume difference between the blank and the baseline design.

$$\text{Material Emission} = 2.57\text{kg } CO_2 + 8.34\text{kg } CO_2 = 10.9\text{kg } CO_2$$

1. Emission Constituent : Product Material Emission

$$\text{Mass of Aluminium 6061 T6 in product} = 0.000107\text{m}^3 \times 2710\text{kg/m}^3 = 0.29\text{kg}$$

$$\text{Product Material Emission} = 0.290\text{kg} \times 8.85\text{kg } CO_2/\text{kg} = 2.57\text{kg } CO_2$$

2. Emission Constituent: : Auxiliary Material Emission

$$\text{Mass of Auxiliary Material} = \text{Mass of Material Waste} = \left[0.000455\text{m}^3 - 0.000107\text{m}^3\right] \times 2710\text{kg/m}^3 = 0.943\text{kg}$$

$$\text{Auxiliary Material Emission} = 0.943\text{kg} \times 8.85\text{kg } CO_2/\text{kg} = 8.34\text{kg } CO_2$$

Emission Category:	:	Fabrication Emission

The computation of the fabrication emission of the baseline design consists of two components: the process emission and the waste emission (Production). For process emissions, the mass of the production waste is used to calculate the energy usage of the milling process, as the production factor reference is measured in units of energy per unit weight removed.

$$\text{Fabrication Emission} = 1.17\text{ kg } CO_2\text{-eq} + 2.48\text{ kg } CO_2\text{-eq} = 3.65\text{ kg } CO_2$$

1. Emission Constituent: : Process Emission

$$\text{Process Emission} = [2.67\text{kWh/kg} \times 0.943\text{kg}] \times 0.465\text{kg } CO_2/\text{kWh} = 1.17\text{kg } CO_2$$

2. Emission Constituent: : Waste Emission

$$\text{Waste Emission (Production)} = 0.943\text{kg} \times 2.63\text{kg } CO_2/\text{kg} = 2.48\text{kg } CO_2$$

Emission Category	:	Design EoL Emission

$$\text{EoL Emission} = \text{Waste Emission (EoL)} = 0.76\text{kg } CO_2$$

1. Emission Constituent	:	Waste Emission (EoL)

$$\text{Waste Emission (EoL)} = 0.29\text{kg} \times 2.63\text{kg } CO_2/\text{kg} = 0.76\text{kg } CO_2$$

Emission Baseline

$$\text{Emission Baseline} = 10.9\text{kg } CO_2 + 3.65\text{kg } CO_2 + 0.76\text{kg } CO_2 = 15.32\text{kg } CO_2$$

that materials in the database must demonstrate design durability against water, and ultraviolet radiation, and have a melting point and maximum service temperature that is higher than 41 °C to be considered for inclusion in the design solution generation. For material durability, given the data qualitative nature of the data from Ansys Granta, a material will be deemed unsuitable if they are labelled for limited use or unacceptable for water durability, or a rating of poor for ultraviolet radiation durability. Table 6 summarises the outcome of the pre-screening process. Out of the initial list of 50 materials, 38 materials fulfil the pre-screening conditions.

4.4. Geometry generation

In the case study, alternative geometries were generated using the generative design feature found in Autodesk Fusion (Autodesk Incb). For the design generation, variations in the starting geometry and manufacturing process were employed to trigger the generation of alternative geometries. Four starting geometries were utilized.

1) Baseline geometry: The original design without any modifications.
2) No starting geometry: No starting geometry is provided except for the preserve region.
3) Modified geometry: Geometry generated by perforating the baseline geometry.
4) Topology optimised geometry: Geometry modified based on topology optimization.

Fig. 7 illustrates the setup for the generative design using the four starting geometries. In the figure, the geometry in red represents obstacle geometry, green indicates preserve geometry, and yellow denotes the starting shape. Obstacle geometry marks regions of the design space to avoid during the design generation process (Autodesk Incc). Preserve geometry denotes features in the design that must be retained in the final design (i.e., the preserve region) while starting shape refers to the initial geometry of the design provided (Autodesk Incc).

Within each setup, three design geometries were generated, each intended for fabrication using either additive, subtractive, or formative

Table 6
Outcome of pre-screening of curated material database.

Idx	Durability		Melting Point, °C	Max Service Temp	Shortlist	Idx	Durability		Melting Point	Max Service Temp	Shortlist
	Water	Ultraviolet Radiation					Water	Ultraviolet Radiation			
M1	Pass	Fail	Pass	Pass	X	M26	Pass	Pass	Pass	Pass	✓
M2	Pass	Fail	Pass	Pass	X	M27	Pass	Pass	Pass	Pass	✓
M3	Pass	Fail	Pass	Pass	X	M28	Pass	Pass	Pass	Pass	✓
M4	Pass	Pass	Pass	Pass	✓	M29	Pass	Pass	Pass	Pass	✓
M5	Pass	Pass	Pass	Pass	✓	M30	Pass	Pass	Pass	Pass	✓
M6	Pass	Pass	Pass	Pass	✓	M31	Pass	Pass	Pass	Pass	✓
M7	Pass	Pass	Pass	Pass	✓	M32	Pass	Pass	Pass	Pass	✓
M8	Pass	Pass	Pass	Pass	✓	M33	Pass	Pass	Pass	Pass	✓
M9	Pass	Pass	Pass	Pass	✓	M34	Pass	Pass	Pass	Pass	✓
M10	Pass	Pass	Pass	Pass	✓	M35	Pass	Pass	Pass	Pass	✓
M11	Pass	Pass	Pass	Pass	✓	M36	Pass	Pass	Pass	Pass	✓
M12	Pass	Pass	Pass	Pass	✓	M37	Pass	Pass	Pass	Pass	✓
M13	Pass	Pass	Pass	Pass	✓	M38	Pass	Pass	Pass	Pass	✓
M14	Pass	Fail	Pass	Pass	X	M39	Pass	Pass	Pass	Pass	✓
M15	Pass	Fail	Pass	Fail	X	M40	Pass	Pass	Pass	Pass	✓
M16	Pass	Pass	Pass	Pass	✓	M41	Pass	Pass	Pass	Pass	✓
M17	Pass	Fail	Pass	Pass	X	M42	Pass	Pass	Pass	Pass	✓
M18	Pass	Fail	Pass	Pass	X	M43	Pass	Pass	Pass	Pass	✓
M19	Pass	Fail	Pass	Pass	X	M44	Pass	Pass	Pass	Pass	✓
M20	Pass	Fail	Pass	Pass	X	M45	Pass	Pass	Pass	Pass	✓
M21	Pass	Fail	Pass	Pass	X	M46	Pass	Pass	Pass	Pass	✓
M22	Pass	Fail	Pass	Pass	X	M47	Pass	Pass	Pass	Pass	✓
M23	Pass	Pass	Pass	Pass	✓	M48	Pass	Pass	Pass	Pass	✓
M24	Pass	Pass	Pass	Pass	✓	M49	Pass	Pass	Pass	Pass	✓
M25	Pass	Pass	Pass	Pass	✓	M50	Fail	Pass	Pass	Pass	X

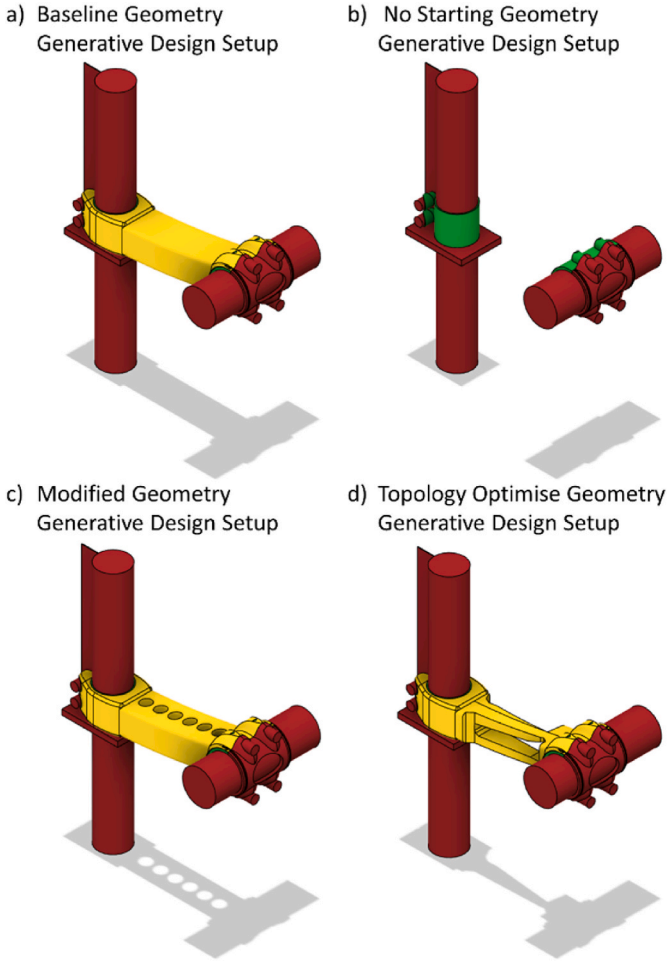


Fig. 7. Generative design setup with four different starting geometry.

manufacturing processes. The generated geometries, along with their respective starting geometries, are compiled in Fig. 8. In the figure, each generated geometry is labelled with a unique geometry identifier below it. This identifier will be used henceforth to refer to their respective generated geometries for the remainder of the chapter. Lastly, the generated geometries are simulated under the testing conditions described in Table 2, and the dynamic baseline is updated accordingly. The results from the simulations as well as the revised material tensile strength requirement for each generated geometry, are summarised in Table 7.

4.5. Geometry-material matching

In the geometry-material matching component, the generated geometries will be matched with the pre-screened materials to create alternative design solutions. For this case study, 12 design geometries and 39 materials (38 pre-screened materials and the baseline material) are considered. The matching process entails comparing each material's tensile stress with the revised required material tensile strength, computed using the required safety factor and the maximum von Mises stress obtained from the simulation of the stage 2 forward bending test, for each generated geometry. A successful match between a material and geometry pair occurs if the tensile strength of the material is higher than the required material tensile strength of the generated geometry. The analysis resulted in a total of 153 design solutions generated from the geometry-material matching. Table 8 summarises the results of this analysis.

4.6. Material-manufacturing process matching

Next, the generated design solutions will be evaluated based on the suitability of the materials to be processed by the assigned manufacturing process. The component functions as a pruning mechanism to remove designs that are not feasible regarding the material's manufacturability. In this case study specific processes were identified for the manufacturing processes considered. For subtractive manufacturing, the material's ability to be processed by fine machining was evaluated. For additive manufacturing, selective laser melting/

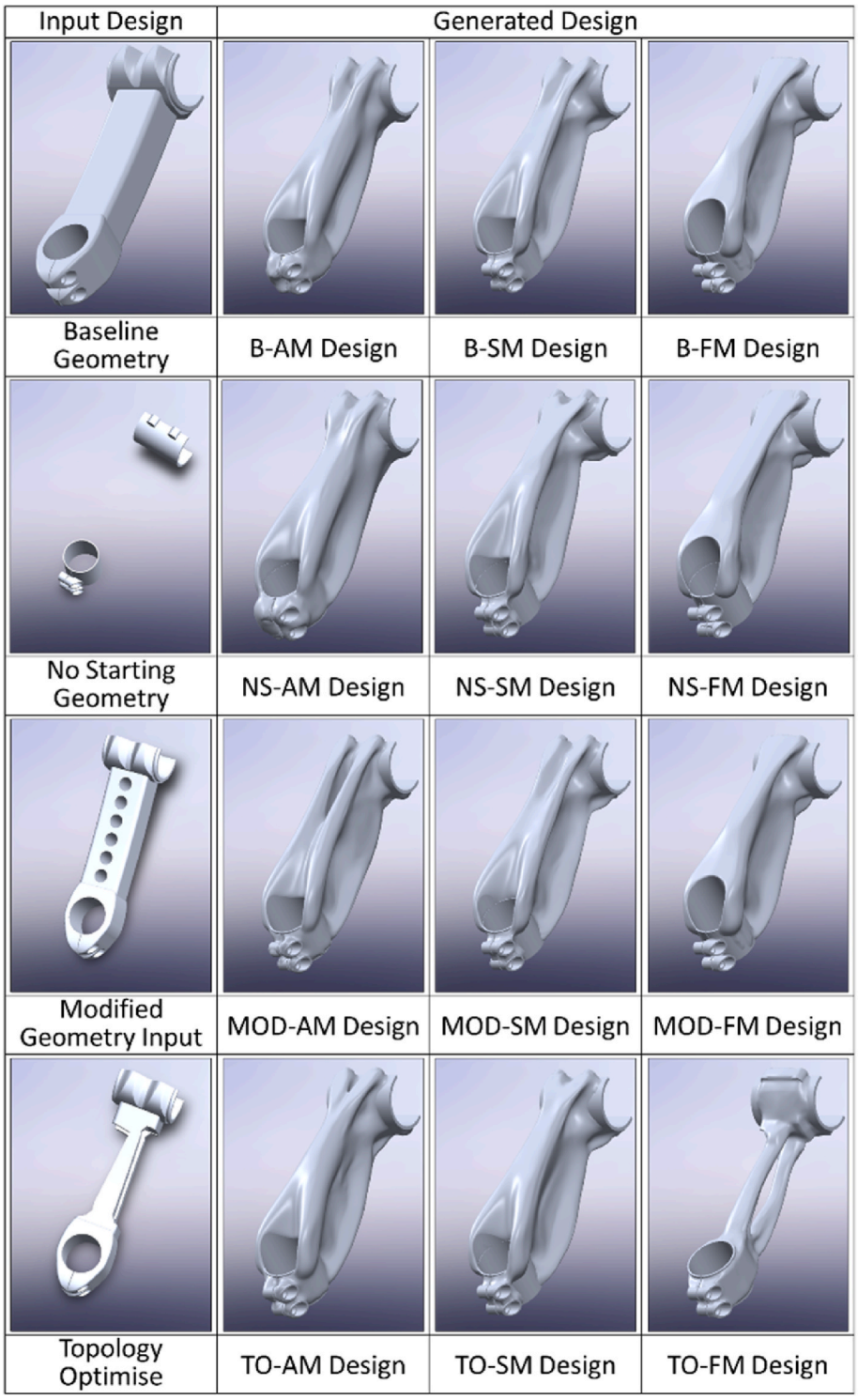


Fig. 8. Compilation of starting geometries and their corresponding generated geometries from generative design.

sintering was considered for metals, while fused filament fabrication was considered for polymers (Papadakis et al., 2018/03). Lastly, for formative manufacturing processes, injection moulding was considered for polymers, and metal casting was considered for metals. The manufacturability of the material for these manufacturing processes is compiled in Appendix D. Table 9 summarises the design solutions that remain after the expanded assessment of the material manufacturability. Following this assessment, the total number of design solutions in contention was reduced to 114.

4.7. Emission pruning

Following the generation of the design solutions, the considered emission constituents of the design solutions will be computed. The emission factors used in the emission computation are extracted from Ansys Granta Selector (I. ANSYS), as detailed in Appendix E. For the EoL emission factor, the carbon footprint from the recycling of the material will be used. For materials that cannot be recycled, the carbon footprint from the combustion of the material will be used instead (Ashby et al.).

Table 7
Summary of simulation results for stage 1 and 2 forward bending tests, along with the updated required tensile strength for the generated geometries.

Design Index	Volume, mm^3	Simulation Outcome			Required Material Tensile Strength, MPa
		Stage 1 Forward Bending Test		Stage 2 Forward Bending Test	
		Maximum von Mises Stress, MPa	Total Displacement, mm	Maximum von Mises Stress, MPa	
B-AM	125500.00	67.42	0.13	109.51	164.27
B-SM	97477.62	160.58	0.21	260.84	391.26
B-FM	95375.35	247.98	0.24	402.83	604.24
NS-AM	141400.00	62.11	0.10	100.88	151.32
NS-SM	89377.19	211.12	0.33	342.93	514.39
NS-FM	89425.24	156.84	0.29	254.77	382.15
MOD-AM	100200.00	120.61	0.23	195.90	293.86
MOD-SM	97447.30	146.92	0.22	238.66	357.99
MOD-FM	100300.00	117.75	0.22	191.27	286.90
TO-AM	109600.00	161.87	0.19	263.04	394.57
TO-SM	104500.00	137.74	0.16	223.85	335.78
TO-FM	45490.36	226.25	1.15	367.68	551.52

While the general equations involved in the computation of the emission constituents remain similar to those used to compute the baseline emission (Table 5), there are deviations in the derivation of certain emission variables for additive and subtractive manufacturing processes.

4.7.1. Additive manufacturing process

In the additive manufacturing processes considered in this case study, apart from the primary material used in the design, additional material is necessary to create support structures. These structures are pivotal for producing intricate geometries, given the layer-by-layer technique intrinsic to additive manufacturing (Jiang et al., 2018). When a layer substantially differs from the preceding one, as with overhanging parts, support structures become indispensable to facilitate the successful manufacturing of these features. When computing the emission of additive manufactured design solutions, emissions associated with these support structures need to be incorporated. Equation (4) describes how the mass of the support structure is being computed in this case study. The computed value will subsequently be used as inputs in equations that require the mass of material waste.

$$\begin{matrix}\text{Mass of}\\ \text{material}\\ \text{waste}\end{matrix} = \begin{matrix}\text{Mass of}\\ \text{support}\\ \text{structure}\end{matrix} = \begin{matrix}\text{Volume}\\ \text{of support}\\ \text{structure}\end{matrix} \times \begin{matrix}\text{Density of}\\ \text{material}\end{matrix} \tag{4}$$

The volume of the required support structure is determined using various software tools tailored for the design geometry earmarked for additive manufacturing. The support structure required for the design solutions is generated using the manufacturing extension of Autodesk Fusion (Autodesk Incd) for metal powder bed fusion (selective laser melting/sintering and Cura (UltiMaker) for the fused filament

Table 8
Outcome of geometry-material matching.

Geometry Index		B-AM	B-SM	B-FM	NS-AM	NS-SM	NS-FM	MOD-AM	MOD-SM	MOD-FM	TO-AM	TO-SM	TO-FM
Required Material Tensile Strength, MPa		164.27	391.26	604.24	151.32	514.39	382.15	293.86	357.99	286.90	394.57	335.78	551.52
Material Index	Tensile Strength, MPa												
MBL	314.00	✓	X	X	✓	X	X	✓	X	✓	X	X	X
M4	45.50	X	X	X	X	X	X	X	X	X	X	X	X
M5	50.45	X	X	X	X	X	X	X	X	X	X	X	X
M6	39.65	X	X	X	X	X	X	X	X	X	X	X	X
M7	60.50	X	X	X	X	X	X	X	X	X	X	X	X
M8	51.75	X	X	X	X	X	X	X	X	X	X	X	X
M9	67.55	X	X	X	X	X	X	X	X	X	X	X	X
M10	67.55	X	X	X	X	X	X	X	X	X	X	X	X
M11	55.50	X	X	X	X	X	X	X	X	X	X	X	X
M12	58.75	X	X	X	X	X	X	X	X	X	X	X	X
M13	28.50	X	X	X	X	X	X	X	X	X	X	X	X
M16	63.00	X	X	X	X	X	X	X	X	X	X	X	X
M23	47.00	X	X	X	X	X	X	X	X	X	X	X	X
M24	42.55	X	X	X	X	X	X	X	X	X	X	X	X
M25	39.90	X	X	X	X	X	X	X	X	X	X	X	X
M26	55.10	X	X	X	X	X	X	X	X	X	X	X	X
M27	46.50	X	X	X	X	X	X	X	X	X	X	X	X
M28	125.00	X	X	X	X	X	X	X	X	X	X	X	X
M29	166.00	✓	X	X	✓	X	X	X	X	X	X	X	X
M30	58.00	X	X	X	X	X	X	X	X	X	X	X	X
M31	290.00	✓	X	X	✓	X	X	X	X	✓	X	X	X
M32	325.50	✓	X	X	✓	X	X	✓	X	✓	X	X	X
M33	632.50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M34	305.00	✓	X	X	✓	X	X	✓	X	✓	X	X	X
M35	315.00	✓	X	X	✓	X	X	✓	X	✓	X	X	X
M36	532.50	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	X
M37	447.50	✓	✓	X	✓	X	✓	✓	✓	✓	✓	✓	X
M38	1397.50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M39	367.50	✓	X	X	✓	X	X	✓	✓	✓	X	✓	X
M40	500.00	✓	✓	X	✓	X	✓	✓	✓	✓	✓	✓	X
M41	105.00	X	X	X	X	X	X	X	X	X	X	X	X
M42	615.00	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M43	1012.50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M44	375.00	✓	X	X	✓	X	X	✓	✓	✓	X	✓	X
M45	350.00	✓	X	X	✓	X	X	✓	X	✓	X	✓	X
M46	510.00	✓	✓	X	✓	X	✓	✓	✓	✓	✓	✓	X
M47	622.50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M48	460.50	✓	✓	X	✓	X	✓	✓	✓	✓	✓	✓	X
M49	475.00	✓	✓	X	✓	X	✓	✓	✓	✓	✓	✓	X

Table 9
Outcome of material-manufacturing process matching.

Material Index	Geometry Index											
	B-AM	B-SM	B-FM	NS-AM	NS-SM	NS-FM	MOD-AM	MOD-SM	MOD-FM	TO-AM	TO-SM	TO-FM
MBL	✓	N.A.	N.A.	✓	N.A.	N.A.	✓	N.A.	X	N.A.	N.A.	N.A.
M29	✓	N.A.	N.A.	✓	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
M31	✓	N.A.	N.A.	✓	N.A.	N.A.	N.A.	N.A.	X	N.A.	N.A.	N.A.
M32	✓	N.A.	N.A.	✓	N.A.	N.A.	✓	N.A.	X	N.A.	N.A.	N.A.
M33	X	✓	X	X	✓	X	X	✓	X	X	✓	X
M34	✓	N.A.	N.A.	✓	N.A.	N.A.	✓	N.A.	✓	N.A.	N.A.	N.A.
M35	✓	N.A.	N.A.	✓	N.A.	N.A.	✓	N.A.	✓	N.A.	N.A.	N.A.
M36	✓	✓	N.A.	✓	✓	✓	✓	✓	✓	✓	✓	N.A.
M37	✓	✓	N.A.	✓	N.A.	✓	✓	✓	✓	✓	✓	N.A.
M38	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
M39	X	N.A.	N.A.	X	N.A.	N.A.	X	✓	✓	N.A.	✓	N.A.
M40	✓	✓	N.A.	✓	N.A.	✓	✓	✓	✓	✓	✓	N.A.
M42	✓	✓	X	✓	✓	X	✓	✓	X	✓	✓	X
M43	✓	✓	X	✓	✓	X	✓	✓	X	✓	✓	X
M44	X	N.A.	N.A.	X	N.A.	N.A.	X	✓	X	N.A.	✓	N.A.
M45	X	N.A.	N.A.	X	N.A.	N.A.	X	N.A.	✓	N.A.	✓	N.A.
M46	X	✓	N.A.	X	N.A.	✓	X	✓	✓	X	✓	N.A.
M47	X	✓	✓	X	✓	✓	X	✓	✓	X	✓	✓
M48	✓	✓	N.A.	✓	N.A.	X	✓	✓	X	✓	✓	N.A.
M49	✓	✓	N.A.	✓	N.A.	✓	✓	✓	✓	✓	✓	N.A.

fabrication process. To streamline computational resources, support structures are exclusively generated for design geometries with at least one designated design solution for the investigated fabrication process. Consequently, only two support structures are generated for the fused filament fabrication process out of the potential four additive manufacturing-assigned geometries. The volume of the generated support structures is compiled in Table 10, while the models of these support structures are documented for review in Appendix F.

Additionally, in this case study, the process emissions for geometries fabricated using additive manufacturing are computed based on the process energy usage calculated with Equation (5). The necessary input for this computation is compiled in Table 10. The fabrication time is derived from simulations conducted using Autodesk Netfabb (Autodesk Ince) for metal powder bed fusion and Cura (UltiMaker) for fused filament fabrication. For the metal powder bed fusion process, the fabrication time for various laser speeds was simulated for each geometry and summarised in Appendix F, with the mid-range fabrication time being selected in the emission computation.

$$\begin{matrix}\text{Energy}\\ \text{Usage}\\ \text{(Additive)}\end{matrix} = \text{Fabrication Time} \times \begin{matrix}\text{Machine}\\ \text{Power}\\ \text{Rating}\end{matrix} \tag{5}$$

4.7.2. Subtractive manufacturing process

In subtractive manufacturing, the fabrication process involves removing material from a material blank to create the desired design geometry. To determine the dimensions of the material blank required, the bounding box of the generated design geometry will be used as a reference. A minimum of 3.175 mm will be added to each dimension of the material blank, this is commonly done for best practice to account for potential variation in the raw material (Xometry).

Equation (6) describes the computation of the material waste generated from the subtractive manufacturing process. Table 11 compiles the dimensions of the material blank and the waste volume that could be generated from the fabrication of the relevant design geometry by subtractive manufacturing.

$$\begin{matrix}\text{Mass of}\\ \text{Material}\\ \text{Waste}\end{matrix} = \left[\begin{matrix}\text{Volume of}\\ \text{material}\\ \text{blank}\end{matrix} - \begin{matrix}\text{Volume of}\\ \text{material}\\ \text{in design}\end{matrix}\right] \times \begin{matrix}\text{Material}\\ \text{Density}\end{matrix} \tag{6}$$

The emission constituent computed for each design solution organised by the design geometry is compiled in Appendix G. Table 12 consolidates the MPE emissions of all shortlisted design solutions. Design solutions with emissions higher than the baseline emissions are highlighted in red, while all viable design solutions that have an emission lower than the baseline emission are highlighted in green. Among the design solutions evaluated, 87 design solutions have emissions lower than the baseline emission and are shortlisted.

4.8. Design shortlisting and post-processing

Following the emission pruning component, the reduction factor of each design solution is computed using Equation (2). The reduction factor of each design solution organised by their rankings can be found in Appendix H. Following the ranking, ten design solutions with the highest reduction factor are selected and simulated under stage 1 and 2 forward bending tests to ensure compliance. Post-processing of the design solutions according to the manufacturing process is not required for this case study, as the generated geometry of the design aligns with the constraints of the manufacturing process assigned to each design

Table 10
Compilation of variables required to compute the process emission required for additive manufactured geometry.

Fabrication Technologies	Metal Powder Bed Fusion		Fused Filament Fabrication	
Machine Power Rating	0.1 kW		0.221 kW	
Geometry Index	Support Volume	Fabrication Time	Support Volume	Fabrication Time
B-AM	2.07 cm^3	55.65 h	6.45 cm^3	24.27 h
NS-AM	2.13 cm^3	61.92 h	6.45 cm^3	43.82 h
MOD-AM	2.26 cm^3	45.75 h	N.A.	N.A.
TO-AM	2.08 cm^3	49.39 h	N.A.	N.A.

Table 11
Summary of material blank and material waste of design geometry fabricated by subtractive manufacturing.

Geometry Index	Dimension of Material Blank			Blank Volume, m^3	Waste Volume, m^3
	Length, mm	Width, mm	Height, mm		
B-SM	162.435	48.175	58.205	4.555E-04	3.580E-04
NS-SM	162.505	48.175	57.435	4.496E-04	3.603E-04
MOD-SM	162.455	48.175	57.875	4.529E-04	3.555E-04
TO-SM	162.635	48.175	57.835	4.531E-04	3.486E-04

Table 12
Compilation of emission for all shortlisted designs.

	Baseline Emission					15.32 kg CO_2						
	Geometry Index											
	B-AM	B-SM	B-FM	NS-AM	NS-SM	NS-FM	MOD-AM	MOD-SM	MOD-FM	TO-AM	TO-SM	TO-FM
MBL	6.55	N.A.	N.A.	7.34	N.A.	N.A.	5.31	N.A.	N.A.	N.A.	N.A.	N.A.
M29	5.76	N.A.	N.A.	8.06	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
M31	6.54	N.A.	N.A.	7.33	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
M32	6.45	N.A.	N.A.	7.23	N.A.	N.A.	5.23	N.A.	N.A.	N.A.	N.A.	N.A.
M33	N.A.	75.32	N.A.	N.A.	74.39	N.A.	N.A.	74.90	N.A.	N.A.	74.89	N.A.
M34	10.44	N.A.	N.A.	11.71	N.A.	N.A.	8.43	N.A.	7.19	N.A.	N.A.	N.A.
M35	10.01	N.A.	N.A.	11.24	N.A.	N.A.	8.09	N.A.	6.86	N.A.	N.A.	N.A.
M36	13.61	40.45	N.A.	15.28	39.95	8.87	10.98	40.22	9.95	11.94	40.22	N.A.
M37	12.53	36.83	N.A.	14.07	N.A.	7.99	10.12	36.63	8.96	11.00	36.62	N.A.
M38	18.02	56.42	12.52	20.25	55.72	11.74	14.52	56.10	13.17	15.81	56.10	5.97
M39	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	16.48	3.79	N.A.	16.46	N.A.
M40	4.66	9.03	N.A.	5.21	N.A.	2.51	3.79	8.97	2.82	4.11	8.95	N.A.
M42	4.67	9.34	N.A.	5.22	9.26	N.A.	3.80	9.28	N.A.	4.12	9.25	N.A.
M43	4.67	10.03	N.A.	5.22	9.95	N.A.	3.80	9.97	N.A.	4.12	9.92	N.A.
M44	N.A.		N.A.	N.A.	N.A.	N.A.	N.A.	11.55	N.A.	N.A.	11.49	N.A.
M45	N.A.		N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	3.17	N.A.	12.54	N.A.
M46	N.A.	11.10	N.A.	N.A.	N.A.	3.01	N.A.	11.04	3.37	N.A.	11.01	N.A.
M47	N.A.	19.45	4.88	N.A.	19.24	4.58	N.A.	19.34	5.13	N.A.	19.31	2.33
M48	5.22	11.16	N.A.	5.84	N.A.	N.A.	4.24	11.10	N.A.	4.60	11.07	N.A.
M49	10.32	28.81	N.A.	11.59	N.A.	6.46	8.34	28.65	7.25	9.07	28.63	N.A.

* The design combination highlighted in red will be pruned as they are higher than the baseline emission

solution. The results from the simulation are compiled in Table 13. From the simulated results, all ten design solutions fulfil the requirements outlined in Table 2, indicating their viability. Therefore, considering the environmental sustainability of the design solutions, the design solution with the combination of the TO-FM geometry and M47 material (Coated steel, stainless steel, terne coated) fabricated by metal casting emerges as the ideal solution for the case study.

To benchmark the performance of the workflow, the highest-ranked design generated from the workflow will be compared with the highest-ranked design produced using generated using generative design for lightweighting in the case study. The implementation of lightweighting considering only the MBL will likely reflect the results shown in the MBL row of Table 12, where the highest emission reduction factor is achieved with the MOD-AM geometry at 65.3%. This contrasts with the 84.8% reduction factor of the optimal alternative design generated using the workflow, highlighting the significant reduction opportunity that is available when the design tri-factor is considered in design generation.

5. Discussion and outlook

5.1. Contribution of work

In this work, a generative ecodesign workflow consisting of seven components was developed. This research yields two significant contributions that are underrepresented in the literature.

Firstly, the work features a functional workflow that integrates ecodesign into generative design. This integration was achieved by incorporating life cycle thinking in different components of the workflow such as the generation of emission ontology in the baseline establishment component and the environmental sustainability evaluation in the emission pruning component of the workflow. While generative design has great potential for ecodesign applications, existing tools often lack the consideration of life cycle thinking, which can compromise their effectiveness in generating environmentally sustainable design solutions. By introducing life cycle thinking into the generative design

Table 13
Summary of simulation results for stage 1 and 2 forward bending tests for the selected design solutions.

Rank	Geometry Index	Material Index	Simulation Outcome				
			Stage 1 Forward Bending Test			Stage 2 Forward Bending Test	
			Displacement, mm	Maximum von Mises Stress, MPa	Factor of Safety	Maximum von Mises Stress, MPa	Factor of Safety
1	TO-FM	M47	0.404	226.152	2.753	367.523	1.694
2	NS-FM	M40	0.097	157.091	3.183	255.172	1.959
3	MOD-FM	M40	0.073	118.168	4.231	191.946	2.605
4	NS-FM	M46	0.095	157.188	3.245	255.331	1.997
5	MOD-FM	M45	0.086	118.444	2.955	192.395	1.819
6	MOD-FM	M46	0.072	118.285	4.312	192.137	2.654
7	MOD-FM	M39	0.188	117.857	3.118	191.441	1.920
8	MOD-AM	M40	0.076	120.089	4.164	195.064	2.563
9	MOD-AM	M42	0.075	119.902	5.129	194.761	3.158
10	MOD-AM	M43	0.075	119.902	8.444	194.761	5.199

process, the proposed workflow brings the consideration of product emissions to the forefront in the consideration of designs generated.

The second contribution of the work is that it facilitates the concurrent consideration of the design tri-factor. As discussed in Appendix A, the design tri-factors consist of interrelated variables that influence one another. Decisions regarding any one of these factors can significantly impact the design options available for the other two. Given the crucial role and influence of the design tri-factor in shaping both the design and environmental sustainability of a mechanical product, considering these factors concurrently expands the design space. This enables the creation and exploration of optimal design solutions that maximise environmental sustainability improvements. Despite the influence of this approach, current ecodesign tools typically address elements of the design tri-factor in isolation, often focusing on only one or two aspects, as highlighted in an auxiliary literature review in Appendix A (Kamalakkannan and Kulatunga, 2021/07; Ferro et al., 2021/01; Pollini and Rognoli, 2021/10; Albiñana and Vila, 2012/10; Tao et al., 2016/01; Bi et al., 2017; Tambouratzis et al., 2014; Zaman et al., 2018/06). Consequently, there has been a lack of ecodesign tools that guide designers to concurrently consider all three elements of the design tri-factor—until the development of this work.

5.2. Future development

In the proposed workflow, a single iteration approach was adopted to generate design solutions aimed at improving the environmental sustainability of the starting design. Even though this approach allows for further design optimization by reapplying the workflow, as discussed in the "Design Shortlisting and Post-Processing" component, the current method for selecting design solutions for iterative processes could be refined. A potential enhancement for future development could involve adopting an evolutionary approach, such as incorporating genetic algorithms, to optimise design inputs during iterative runs of the workflow. Fig. 9 illustrates the revised workflow incorporating this iterative process.

In the revised generative ecodesign workflow, the initial run follows the original sequence, progressing from the "Baseline Establishment" component to the "Design Shortlisting Post-Processing" component. After completing the initial sequence, the "Genetic Algorithm" component utilizes the top design solutions to create new design tri-factor combinations. These combinations are then used as inputs for a second workflow iteration, starting from the "Geometry Generation" components. The resulting design geometries are processed through the subsequent workflow stages, concluding again with the "Design Shortlisting and Post-Processing" component. If further iterations are required, the top designs from each iteration are fed back into the "Genetic Algorithm" component, and the workflow restarts from the "Geometry Generation" component. This iterative loop continues until the solution converges, with the sequence represented by the red dotted arrow in Fig. 9.

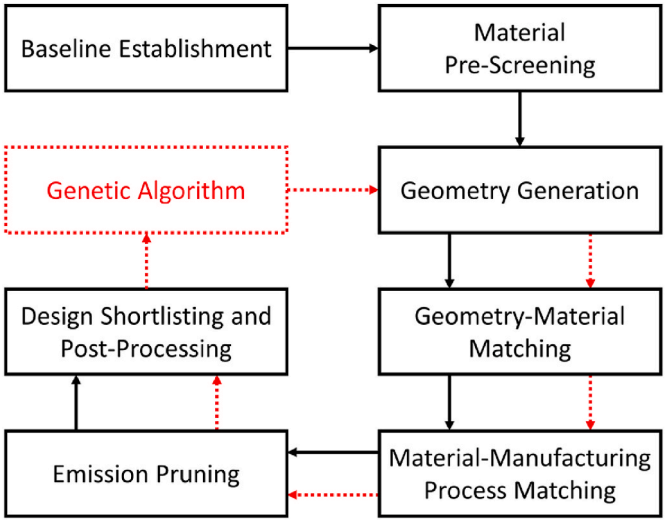


Fig. 9. Revised generative ecodesign workflow for iterative operation. The red dotted lines imply the sequence of events after the initial run of the workflow.

Another future development would involve supporting the consideration of new technologies in the workflow. As newer manufacturing processes and materials emerge, designers will likely explore design solutions that leverage these innovations. However, a key challenge in integrating new technologies is the lack of data to model their environmental impact. To better support designers, the workflow could incorporate prospective LCA capabilities such as using patent search (Spreafico et al., 2023/06), to estimate emissions associated with these new technologies.

6. Conclusion

To address the challenges of our times, the manufacturing industry requires innovative design solutions that can significantly reduce emissions associated with production. This necessitates designers to push beyond the boundaries of their imagination and expertise in generating design solutions. For this to occur, ecodesign tools must evolve from prescriptive approaches to generative ones —a shift that is currently underrepresented in the literature. Recognizing this gap, a generative ecodesign workflow consisting of seven components was developed in this work.

This workflow integrates life cycle thinking with generative design to generate, analyse, and evaluate alternative design solutions aimed at enhancing the environmental sustainability of the starting design. A key significance of this work is that it represents the first reported instance of incorporating ecodesign into generative design while concurrently considering the design tri-factor.

To demonstrate its application, the workflow was implemented in a case study involving the design tri-factor of a mountain bicycle's handlebar stem. In addition to illustrating the workflow's implementation, the case study highlights the importance of concurrently considering the design tri-factor to maximise the environmental sustainability of design solutions. In performance benchmarking, the reduction factors achieved by the best design generated using generative design considering the design tri-factor and current lightweighting practices were 84.8% and 65.3% respectively. While this discrepancy is not always present, it underscores the potential improvement opportunities missed when the design tri-factor is not concurrently considered. This further emphasizes the value of this work in helping designers generate design solutions that maximise environmental sustainability.

CRediT authorship contribution statement

Amos Wei Lun Lee: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kevin Kai Wern Seah:** Methodology, Investigation, Data curation. **Bing Feng Ng:** Supervision, Resources, Project administration. **Ee Teng Zhang:** Supervision. **Wen Feng Lu:** Supervision. **Jonathan Sze Choong Low:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clet.2025.100890.

Data availability

Data will be made available on request.

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